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ORIGINAL PAPER

The role of ethnic enclaves and neighborhood socioeconomic status in invasive breast cancer incidence rates among Asian American, Native Hawaiian, and Pacific Islander females in California

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Abstract

Purpose Few studies have examined whether the incidence rates of invasive breast cancer among Asian American, Native Hawaiian, and Pacific Islander (AANHPI) populations differ by the neighborhood social environment. Thus, we examined associations of ethnic enclave and neighborhood socioeconomic status (nSES) with breast cancer incidence rates among AANHPI females in California.

Methods A total of 14,738 AANHPI females diagnosed with invasive breast cancer in 2008–2012 were identified from the California Cancer Registry. AANHPI ethnic enclaves (culturally distinct neighborhoods) and nSES were assessed at the census tract level using 2007–2011 American Community Survey data. Breast cancer age-adjusted incidence rates and incidence rate ratios (IRRs) were estimated for AANHPI ethnic enclave, nSES, and their joint effects. Subgroup analyses were conducted by stage of disease.

Results The incidence rate of breast cancer among AANHPI females living in lowest ethnic enclave neighborhoods (quintile (Q)1) were 1.21 times (95% Confidence Interval (CI) 1.11, 1.32) that of AANHPI females living highest ethnic enclave neighborhoods (Q5). In addition, AANHPI females living in highest vs. lowest SES neighborhoods had higher incidence rates of breast cancer (Q5 vs. Q1 IRR = 1.30, 95% CI 1.22 to 1.40). The incidence rate of breast cancer among AANHPI females living in low ethnic enclave + high SES neighborhoods was 1.32 times (95% CI 1.25, 1.39) that of AANHPI females living in high ethnic enclave + low SES neighborhoods. Similar patterns of associations were observed for localized and advanced stage disease.

Conclusion For AANHPI females in California, incidence rates of breast cancer differed by nSES, ethnic enclave, when considered independently and jointly. Future studies should examine whether the impact of these neighborhood-level factors on breast cancer incidence rates differ across specific AANHPI ethnic groups and investigate the pathways through which they contribute to breast cancer incidence.

Keywords Breast cancer incidence · Asian American, Native Hawaiian, and Pacific Islander (AANHPI) · Neighborhood socioeconomic status · Ethnic enclave · Social environment

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Introduction

For females in the U.S., breast cancer is the most commonly diagnosed cancer. Over 297,790 new cases of invasive breast cancer were estimated to be diagnosed in women in 2023 [1] and projected to reach 364,000 by 2040 [2]. In the U.S., Asian American (AA) populations are one of the fastest-growing racial and ethnic groups with the largest population of AA in the U.S. residing in California [3, 4]. Of note, among AA females in the U.S., breast cancer incidence increased at a rate of 1.17% per year from 1988 to 2013 [5]. Additionally, Native Hawaiian and Pacific Islander (NHPI) females experience a disproportionate burden of breast cancer compared to other Asian ethnic groups [6, 7].

Neighborhood contextual factors are increasingly recognized as risk factors for breast cancer incidence [8]. Among AANHPI females an increased incidence of breast cancer has been reported among those residing in neighborhoods of higher socioeconomic status [9] and/or areas that are less culturally distinct [8]. Ethnic enclaves are neighborhoods that are more culturally distinct, typically defined by higher concentrations of specific racial and ethnic groups, recent immigrants, and with households that have limited English-proficiency or are linguistically isolated [10]. Moreover, ethnic enclaves tend to have businesses and social communities that reflect the cultural values of their residents. Although ethnic enclaves can offer valuable social support that positively impact health, they are often underserved, resulting in a lack of necessary resources for sustaining a healthy lifestyle [11]. However, in recent literature, researchers have found that AANHPI enclaves in California typically are not of lower SES and are more likely to have greater access to primary healthcare [12].

Neighborhood socioeconomic status (nSES) has been reported to be an independent risk factor for breast cancer [9, 13]. A positive association (i.e., higher incidence among those residing in higher SES neighborhoods) has been reported in several studies, including studies of AANHPI females [8, 9, 13, 14]. While the association between nSES and breast cancer incidence is well established, there remains a gap in our knowledge of whether living in AANHPI ethnic enclaves may influence this association [9, 13, 14]. Thus, this study investigated the separate and joint effect of ethnic enclave and nSES on invasive breast cancer incidence in a population-based cross-sectional study of AANHPI females diagnosed with invasive breast cancer from 2008 to 2012 in California. Additionally, subgroup analyses by stage of disease were conducted to examine possible differences in the patterns of association between localized vs. advanced disease.

Methods

Study population and data

Using data from the California Cancer Registry (CCR), part of the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) program, we obtained information on newly diagnosed invasive breast cancer cases (Internal Classification of Diseases for Oncology, 3rd Edition (ICD-O-3) site code C50.0-C50.9, histology codes 8000-8700, 8720-8790, and 8982-8983) from January 1, 2008 to December 31, 2012. Registry data are routinely derived from the medical record on age at diagnosis, race and ethnicity, and stage of diagnosis. SEER summary stage at diagnosis was defined as local (1 = localized), regional (2 = regional by direct extension; 3 = regional by lymph nodes; 4 = regional by direct extension and lymph nodes; 5 = regional not otherwise specified), remote (7 = remote), and unknown (8 = not abstracted; 9 = unknown or not specified). Regional and remote stage were combined as advanced disease.

A total of 14,738 female AANHPI invasive breast cancer cases diagnosed from 2008 to 2012, who were at least 20 years of age were included in this study after excluding females who had unknown breast cancer stage ($n = 192$). Five-year cumulative incidence rates were calculated using 2008–2012 case counts (numerator) and 2010 Census population counts (denominator) at the census-tract level. The age, sex, and race-specific population counts from the 2010 Census were multiplied by five as intercensal population counts were not available. Census tracts were assigned to breast cancer cases based on residential address at diagnosis to append neighborhood data.

Ethnic enclave and neighborhood socioeconomic status

For this study, a neighborhood was defined using census tracts, a geographic unit with approximately 4,000 residents on average in California that has been used in many studies of neighborhood health research [15]. AANHPI ethnic enclave was defined as neighborhoods with a high concentration of residents with self-reported AANHPI race, recent immigration, AANHPI language use, and linguistic isolation [16] compared to other neighborhoods [10]. A principal component analysis was conducted as previously described [10, 17, 18] using 2007–2011 American Community Survey (ACS) data with four variables capturing the domains described above. Neighborhood socioeconomic status (nSES) was assessed using an established index [16, 19] based on information derived from 2007 to 2011 ACS data on education, household income, income relative to poverty line, occupation, employment status, rent, and house value [19]. For both ethnic enclave (quintile (Q) 5 = most culturally distinct neighborhood, Q1 = least culturally distinct neighborhood) and nSES (Q5 = highest nSES, Q1 = lowest nSES), census tracts were assigned the appropriate composite score, and then categorized into quintiles based on the statewide distribution of scores across all census tracts for the decennial Census year of 2010.

A joint measure of ethnic enclave and nSES was developed using the dichotomous ethnic enclave and nSES measures (i.e., low enclave [Q1-3], and high enclave [Qs 4 and 5), low nSES [Q1-3], high nSES [Qs 4 and 5]). The joint variable included four categories to describe neighborhoods: low enclave + low nSES, low enclave + high nSES, high enclave + low nSES, high enclave + high nSES.

Statistical analysis

We calculated frequencies and relative distributions for age group (20–29, 30–39, 40–49, 50–59, 60–69, 70–79, & 80 +), year of diagnosis, summary stage (localized, regional/distant), ethnic enclave, and nSES. Age-adjusted incidence rates (AAIR) were per 100,000 were age-adjusted using the 2000 U.S. standard population. To examine differences in incidence rates by nSES and ethnic enclave, we calculated incidence rate ratios (IRR) for ethnic enclave and nSES quintiles. Each quintile was compared to a reference (i.e., ethnic enclave (Q5) and nSES (Q1)) and then assessed for trends. To examine differences in joint ethnic enclave + nSES, we calculated IRRs among low enclave + high nSES, high enclave + high nSES, and low enclave + low nSES compared to high enclave + low nSES neighborhoods. High enclave + low nSES was selected as the reference group as each of these categories had the lowest incidence rate of invasive breast cancer for AANHPI females. We used the NCI's SEER*Stat software (version 8.4.2) to compute AAIR per 100,000 population and IRR and 95% confidence intervals (Tiwari modification) [20, 21]. Subgroup analyses were conducted for localized and advanced disease. All statistical tests conducted were two-sided, with a $P < 0.05$ indicating statistical significance. P values for trend analysis were conducted using SAS (version 9.4). Relevant data are available through the California Cancer Registry.

Results

Cancer case characteristics

Table 1 presents frequencies and distributions of 14,738 AANHPI California females with invasive breast cancer diagnosed from 2008 to 2012. Of them, 71% were diagnosed at ages of 50 or older, 65% with localized disease, and 34% with advanced disease (Table 1). Approximately 74% of AANHPI breast cancer cases resided in more culturally distinct neighborhoods (AANHPI enclave Q4 and Q5). The majority of AANHPI breast cancer cases (57%) resided in high nSES neighborhoods (Q4 and Q5) while 43% resided in low SES neighborhoods (Q1-Q3).

For AANHPI females, AAIR decreased with higher ethnic enclave quintiles such that females residing in the least culturally distinct neighborhoods had the highest incidence rate of breast cancer (Q1 AAIR: 115.10 per 100,000; 95% CI: 105.57, 125.32) while those living in highest AANHPI ethnic enclave neighborhoods (more culturally distinct) had the lowest incidence rate (Q5 IR: 95.21; 95% CI: 93.07, 97.39) (Table 2). Relative to AANHPI females residing in the highest ethnic enclave neighborhoods, females residing in the lowest ethnic enclave had a 21%

Table 1 Study Characteristics of Female Asian American/Native Hawaiian/Pacific Islander Invasive Breast Cancer Cases Reported in California from 2008 to 2012 (*n* = 14,738)

	n	%
Age at Diagnosis (years)		
20–29	101	0.68
30–39	947	6.4
40–49	3240	22
50–59	4095	28
60–69	3493	24
70–79	1944	13
80 +	918	6.2
Year of Diagnosis		
2008	2702	18
2009	2766	19
2010	2945	20
2011	3097	21
2012	3228	22
SEER Summary Stage*		
Localized	9586	65
Regional/Distant	4960	34
Ethnic Enclave		
Q1 Low Enclave	557	4.0
Q2 Low Enclave	1273	8.0
Q3 Low Enclave	2106	14
Q4 High Enclave	3184	22
Q5 High Enclave	7618	52
Neighborhood SES		
Q1 Low Neighborhood SES	1134	8.0
Q2 Low Neighborhood SES	2052	14
Q3 Low Neighborhood SES	3167	21
Q4 High Neighborhood SES	3872	26
Q5 High Neighborhood SES	4513	31

*Unknown stage (*n*) = 192 (1.31%)

Table 2 Invasive Breast Cancer Age-Adjusted Incidence Rates (AAIR) and Incidence Rate Ratios (IRR) by Ethnic Enclaves for Asian American/Native Hawaiian/Pacific Islander Females Reported in California from 2008 to 2012

Population	Cases (n)	AAIR[a] (95% CI)	IRR (95% CI)
7,136,940	7618	95.21 (93.07–97.39)	(ref)
2,752,040	3184	104.50 (100.86–108.24)*	1.09 (1.05–1.14)
1,647,505	2106	112.54 (107.72–117.52)*	1.18 (1.13–1.24)
969,415	1273	113.36 (107.13–119.87)*	1.19 (1.12–1.26)
383,450	557	115.10 (105.57–125.32)*	1.21 (1.11–1.32)
			0.01**

[a]Age adjusted incidence rates (AAIR) per 100,000 using the 2000 U.S. standard population (18 age groups-Census P25-1130); 95% Confidence Intervals (CI)

*Indicates statistical significance ($p < 0.05$) comparing to high ethnic enclave

**Indicates statistical significance ($p < 0.05$) of a trend

higher incidence rate of invasive breast cancer (95% CI: 1.11, 1.32). Residing in neighborhoods that were less culturally distinct was associated with increasing incidence of breast cancer ($P_{trend} = 0.01$).

Furthermore, the AAIR increased with nSES such that AANHPI females residing in the lowest SES neighborhoods (Q1 AAIR: 83.59; 95% CI: 78.67, 88.73) had the lowest incidence rate while those residing in the highest SES neighborhoods had the highest incidence rate of breast cancer (Q5 AAIR: 109.20 per 100,000; 95% CI: 105.97, 112.52) (Table 3). Relative to those residing in the lowest SES neighborhoods, AANHPI females residing in the highest SES neighborhoods had a 30% higher incidence rate of invasive breast cancer (IRR = 1.30; 95% CI: 1.22, 1.40). Residing in neighborhoods of higher SES was associated with an increasing incidence of breast cancer ($P_{trend} = 0.01$).

Table 4 shows AAIR and IRR of breast cancer for each category of the joint variable compared to in the reference group, high enclave + low nSES. AANHPI females living in neighborhoods of low ethnic enclave + high nSES (IRR = 1.32 95% 1.25–1.39), low ethnic enclave + low nSES (IRR = 1.23; 95% 1.16–1.29), and high ethnic enclave + high nSES (IRR = 1.20; 95% 1.15–1.24) had a statistically significantly higher incidence rate of breast cancer compared to females living in high ethnic enclave + low nSES neighborhoods.

In a subgroup analysis by stage of disease, AANHPI females with localized stage at diagnosis residing the least

Table 3 Invasive Breast Cancer Age-Adjusted Incidence Rates (AAIR)[a] and Incidence Rate Ratios (IRR) by neighborhood SES (nSES) for Asian American/Native Hawaiian/Pacific Islander Females Reported in California from 2008 to 2012

nSES	Population	Cases (*n*)	AAIR[a] (95% CI)	IRR (95% CI)
Q1 Low	1,209,465	1,134	83.59 (78.67–88.73)	(ref)
Q2	1,948,800	2,052	90.18 (86.26–94.24)*	1.08 (1.00–1.16)
Q3	2,752,600	3,167	100.54 (97.02–104.15)*	1.20 (1.12–1.29)
Q4	3,220,080	3,872	107.88 (104.47–111.38)*	1.39 (1.21–1.38)
Q5 High	3,758,570	4,513	109.20 (105.97–112.52)*	1.30 (1.22–1.40)
p-trend				0.00800**

[a]Age adjusted incidence rates (AAIR) per 100,000 using the 2000 U.S. standard population (18 age groups-Census P25-1130); 95% Confidence Intervals (CI)

*Indicates statistical significance ($p < 0.05$) comparing to low nSES

**Indicates statistical significance ($p < 0.05$) of a trend

Table 4 Invasive Breast Cancer Incidence Rate Ratios[a] by Ethnic Enclave and Neighborhood SES for Asian American/Native Hawaiian/ Pacific Islander females reported in California from 2008 to 2012

	n	*n*	AAIR (95% CI)	IRR (95% CI)
High Ethnic Enclave & Low nSES	4,437,330	4,532	88.63 (86.03–91.30)	(ref)
High Ethnic Enclave & High nSES	5,451,650	6,270	106.09 (103.44–108.79)*	1.20 (1.15–1.24)
Low Ethnic Enclave & Low nSES	1,473,460	1,821	108.95 (103.93–114.16)*	1.23 (1.16–1.29)
Low Ethnic Enclave & High nSES	1,526,910	2,115	116.70 (111.69–121.89)*	1.32 (1.25–1.39)

[a]Age adjusted incidence rates (AAIR) per 100,000 using the 2000 U.S. standard population (18 age groups-Census P25-1130); 95% Confidence Intervals (CI)

[b]Low Enclave (Q1 through 3) and High Enclave (Q4 & 5)

[c]Low nSES (Q1 through 3) and High nSES (Q4 & 5)

*Indicates statistical significance ($p < 0.05$) comparing to high ethnic enclave & low nSES culturally distinct neighborhoods (least culturally distinct) had the highest incidence rate of breast cancer (Q1 AAIR: 75.16 per 100,000; 95% CI: 67.55, 83.46) while those living in highest AANHPI ethnic enclave neighborhoods (more culturally distinct) had the lowest incidence rate (Q5 IR: 61.78; 95% CI: 60.05, 63.54) (Table S1). In contrast, AANHPI females with advanced stage at diagnosis residing in the second to lowest AANHPI ethnic enclave neighborhood had the highest incidence rate of breast cancer (Q2 AAIR: 38.24 per 100,000; 95% CI: 34.67, 42.09) (Table S1). Moreover, AANHPI females living in highest vs. lowest SES neighborhoods had higher incidence rates of localized (Q5 vs. Q1 IRR = 1.44, 95% CI 1.32–1.57) and advanced stage (Q5 vs. Q1 IRR = 1.12, 95% CI 1.00–1.25) breast cancer (Table S2). Similar associations were observed for joint variable of ethnic enclave and nSES when comparing AANHPI females that resided in low ethnic enclave + high nSES neighborhoods to AANHPI females that reside in high ethnic enclave + low nSES neighborhoods for localized stage (IRR = 1.40; 95% 1.31–1.50) and advanced stage at diagnosis (IRR = 1.18; 95% 1.07–1.29) (Table S3).

Discussion

For AANHPI females in California, residing in less culturally distinct neighborhoods (low ethnic enclaves) was associated with higher incidence rates of invasive breast cancer. In addition, AANHPI females living in neighborhoods of higher SES experienced higher incidence rates of invasive breast cancer compared to those residing in neighborhoods of lower SES. When ethnic enclave and nSES were assessed jointly, the association of ethnic enclave status and invasive breast cancer incidence among AANHPI females in California seemed to be driven by nSES, i.e., AANHPI females residing in neighborhoods of higher SES experience higher incidence rates of breast cancer compared to AANHPI residing in neighborhoods of lower SES, regardless of ethnic enclave status.

In prior studies, the examination of AANHPI ethnic enclave and cancer has focused on incidence rates of cancers of lung, liver, hepatocellular carcinoma, colorectal, Hodgkin lymphoma, and non-Hodgkin lymphoma [10, 17, 22–25]. These studies report similar patterns of associations as we observed for residing in AANHPI ethnic enclave and breast cancer incidence whereby individuals residing in areas that were less culturally distinct (low ethnic enclave) experienced higher incidence of cancer [22–25]. Our findings of increased incidence of breast cancer among AANHPI females residing in higher SES neighborhoods are consistent with previous studies of breast cancer [8, 13, 14]. Also, our findings that residing in neighborhoods that were less culturally distinct (low enclave) and higher SES experienced higher incidence rates of breast cancer compared neighborhoods that are were culturally distinct (high ethnic enclave) and lower SES are consistent with prior studies of these neighborhood-level factors for other cancer sites [23–25]. It is plausible that the observed higher breast cancer incidence for AANHPI females residing in low enclave + high SES neighborhoods is driven primarily by nSES, as we also see similarly higher breast cancer incidence for females residing in high enclave + high SES neighborhoods. Prior studies have reported a higher incidence of breast cancer among females residing in high SES neighborhoods [13, 19, 26] that may be related to higher proportion of residents in those neighborhoods with established breast cancer risk factors, including lower parity, delayed age at first birth, later age at menopause, and greater access to healthcare resources including breast cancer screening.

The disparities observed in invasive breast cancer incidence by ethnic enclave status and nSES are likely influenced by additional neighborhood and individual-level factors. Characteristics of the neighborhood such as density and access to social support and healthcare are plausible pathways through which residence in AANHPI enclaves can contribute to breast cancer risk. In a recent study conducted by Guan et al. [12], researchers investigated the geographic accessibility of healthcare by ethnic enclave for Asian Americans across five states, including California, and found that ethnic enclaves tend to have less poverty, fewer uninsured individuals, and higher primary care accessibility compared to neighborhoods that are not enclaves. Thus, such characteristics of the neighborhood may be a plausible pathway by which AANHPI females residing in higher ethnic enclave experience lower incidence rates of invasive breast cancer in comparison to those residing in neighborhoods of lower ethnic enclave. Additionally, individual health behaviors such as nutrition, smoking prevalence, and alcohol consumption may be influenced by neighborhood environments such as AANHPI enclaves and may contribute to breast cancer risk. In a study conducted by Kandula et al., [27] researchers examined smoking prevalence among Asian Americans by neighborhood factors and found a higher smoking prevalence for Asian American females who reside in areas with a lower concentration of Asian residents [27]. Future studies are needed to better understand how neighborhood characteristics may contribute to breast cancer risk so that they can better inform clinical- and community-level interventions. For example, providing culturally relevant and in-language health education resources for AANHPI females can be prioritized in neighborhoods with higher risk.

This study has several strengths. Our large population-based sample of invasive breast cancer cases allowed for sufficient numbers to evaluate breast cancer rates among AANHPI females by ethnic enclave and nSES. Additionally, we examined established measures of ethnic enclave and nSES [10, 19, 28, 29]. However, we note some limitations of this study. As a cross-sectional study, we acknowledge that cancer incidence may not align with the critical window of neighborhood exposure. We were unable to disaggregate the AANHPI population into distinct Asian ethnic group due to lack of reliable denominator data for the most recent years of diagnosis. Lastly, we were unable to examine individual-level breast cancer risk factors such as smoking, reproductive factors, overweight/obesity, physical activity, and alcohol use. It is important to note that this work can be updated in the future using 2020 Census denominator data as well as appropriate data from the ACS for comparison to our findings.

Conclusion

Our study indicates that AANHPI ethnic enclaves and nSES jointly influence breast cancer risk among AANHPI females in California and the association appears to be driven largely by nSES. Future research should evaluate these neighborhood factors across distinct Asian American ethnic groups. Multilevel, longitudinal data are needed to help elucidate the pathways through which these neighborhood factors jointly influence risk.

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Data availability Data/SEER*Stat sessions are available upon request.

Declarations

Competing interest The authors declare no competing interests.

References

1. National Breast Cancer Foundation, INC. https://www.nationalbreastcancer.org/breast-cancer-facts/
2. Arzanova E, Mayorovitz H (1998) The epidemiology of breast cancer. Imaging 10(1):4. https://doi.org/10.36255/exon-publications-breast-cancer-epidemiology
3. Brown A. U.S. Hispanic and Asian populations growing, but for different reasons. Pew Research Center. Published 2014. https://www.pewresearch.org/short-reads/2014/06/26/u-s-hispanic-and-asian-populations-growing-but-for-different-reasons/
4. Von Behren J, Abrahão R, Goldberg D, Gomez SL, Setiawan VW, Cheng I (2018) The influence of neighborhood socioeconomic status and ethnic enclave on endometrial cancer mortality among Hispanics and Asian Americans/Pacific Islanders in California. Cancer Causes Control 29(9):875–881. https://doi.org/10.1007/s10552-018-1063-7
5. Gomez SL, Von Behren J, McKinley M et al (2017) Breast cancer in Asian Americans in California, 1988–2013: increasing incidence trends and recent data on breast cancer subtypes. Breast Cancer Res Treat 164(1):139–147. https://doi.org/10.1007/s10549-017-4229-1
6. Jemal A, Siegel RL (2015) Cancer statistucs for Asian Americans, Native Hawaiians, and Pacific Islanders, 2015: convergence of incidence between males and females. CA Cancer J Clin 2(17):623–624. https://doi.org/10.3322/caac.21335.Cancer
7. CAL*Explorer: An interactive website for California Cancer Registry (CCR) cancer statistics [Internet]. The CCR is a program of the California Department of Public Health. Published 2023. Accessed March 17, 2024. https://explorer.ccrcal.org/application.html?site=55&data_type=7&graph_type=10&compareBy=asian_ethnicity&chk_asian_ethnicity_10=10&chk_asian_ethnicity_11=11&chk_asian_ethnicity_12=12&chk_asian_ethnicity_13=13&chk_asian_ethnicity_14=14&chk_asian_ethnicity_
8. Morey BN, Gee GC, Wang MC et al (2022) Neighborhood contexts and breast cancer among Asian American Women. J Immigr Minor Heal 24(2):445–454. https://doi.org/10.1007/s10903-021-01196-6
9. Krieger N Jr, Quesenberry C, Peng T et al (1999) Social class, race/ethnicity, and incidence of breast, cervix, colon, lung, and prostate cancer among Asian, black, Hispanic, and white residents of the San Francisco Bay Area, 1988±92 (United States) Nancy. Cancer Causes Control 10:525–537
10. Sangaramoorthy M, Yang J, Guan A et al (2022) Asian American/Pacific Islander and Hispanic ethnic enclaves, Neighborhood socioeconomic status, and hepatocellular carcinoma incidence in California: an update. Cancer Epidemiol Biomarkers Prev 31(2):382–392. https://doi.org/10.1158/1055-9965.EPI-21-1035
11. Bécares L, Shaw R, Nazroo J et al (2012) Ethnic density effects on physical morbidity, mortality, and health behaviors: a systematic

review of the literature. Am J Public Health 102(12):33–66. https://doi.org/10.2105/AJPH.2012.300832
12. Guan A, Pruitt SL, Henry KA et al (2023) Asian American enclaves and healthcare accessibility: an ecologic study across five states. Am J Prev Med 65(6):1015–1025. https://doi.org/10.1016/j.amepre.2023.07.001
13. Liu L, Deapen D, Bernstein L (1998) Socioeconomic status and cancers of the female breast and reproductive organs: A comparison across racial/ethnic populations in Los Angeles County, California (United states). Cancer Causes Control 9(4):369–380. https://doi.org/10.1023/A:1008811432436
14. Reynolds P, Hurley SE, Quach AT et al (2005) Regional variations in breast cancer incidence among California women, 1988–1997. Cancer Causes Control 16(2):139–150. https://doi.org/10.1007/s10552-004-2616-5
15. Krieger N, Chen JT, Waterman PD, Soobader MJ, Subramanian SV, Carson R (2002) Geocoding and monitoring of US socioeconomic inequalities in mortality and cancer incidence: does the choice of area-based measure and geographic level matter? The public health disparities geocoding project. Am J Epidemiol 156(5):471–482. https://doi.org/10.1093/aje/kwf068
16. Yang J, Schupp CW, Harrati A, Clarke C, Keegan TH, Gomez SL. Developing an area based socioeconomic measure from American Community Survey data. Cancer Prevention Institute of California. Accessed 24 Feb 2022. https://cancerregistry.ucsf.edu/sites/g/files/tkssra1781/f/wysiwyg/Yang%20et%20al.%202014_CPIC_ACS_SES_Index_Documentation_3-10-2014.pdf
17. Chang ET, Yang J, Alfaro-Velcamp T, So SKS, Glaser SL, Gomez SL (2010) Disparities in liver cancer incidence by nativity, acculturation, and socioeconomic status in California Hispanics and Asians. Cancer Epidemiol Biomarkers Prev 19(12):3106–3118. https://doi.org/10.1158/1055-9965.EPI-10-0863
18. Gomez SL, Glaser SL, Mcclure LA (2012) Incidence and outcomes in populations. Cancer Causes Control 22(4):631–647. https://doi.org/10.1007/s10552-011-9736-5.The
19. Yost K, Perkins C, Cohen R, Morris C, Wright W (2001) Socioeconomic status and breast cancer incidence in California for different race/ethnic groups. Cancer Causes Control 12(8):703–711. https://doi.org/10.1023/A:1011240019516
20. Surveillance Research Program, National Cancer Institute SEER*Stat software (seer.cancer.gov/seerstat) version 8.4.2. https://seer.cancer.gov/help/seerstat/data/citation-of-software-and-data-source
21. Tiwari RC, Clegg LX, Zou Z (2006) Efficient interval estimation for age-adjusted cancer rates. Stat Methods Med Res 15(6):547–569. https://doi.org/10.1177/0962280206070621
22. DeRouen MC, Hu L, McKinley M et al (2018) Incidence of lung cancer histologic cell-types according to neighborhood factors: a population based study in California. PLoS ONE 13(5):1–15. https://doi.org/10.1371/journal.pone.0197146
23. Ladabaum U, Clarke CA, Press DJ et al (2014) Colorectal cancer incidence in asian populations in california: effect of nativity and neighborhood-level factors. Am J Gastroenterol 109(4):579–588. https://doi.org/10.1038/ajg.2013.488
24. Glaser SL, Chang ET, Clarke CA, Keegan THM, Yang J, Gomez SL (2015) Hodgkin lymphoma incidence in ethnic enclaves in California. Leuk Lymphoma 56(12):3270–3280. https://doi.org/10.3109/10428194.2015.1026815
25. Clarke CA, Glaser SL, Gomez SL et al (2011) Lymphoid malignancies in U.S. Asians: incidence rate differences by birthplace and acculturation. Cancer Epidemiol Biomarkers Prev 20(6):1064–1077. https://doi.org/10.1158/1055-9965.EPI-11-0038
26. Akinyemiju TF, Genkinger JM, Farhat M, Wilson A, Gary-Webb TL, Tehranifar P (2015) Residential environment and breast cancer incidence and mortality: a systematic review and meta-analysis. BMC Cancer. https://doi.org/10.1186/s12885-015-1098-z
27. Kandula NR, Wen M, Jacobs EA, Lauderdale DS (2009) Association between neighborhood context and smoking prevalence among Asian Americans. Am J Public Health 99(5):885–892. https://doi.org/10.2105/AJPH.2007.131854
28. Froment M-A, Gomez SL, Roux A, DeRouen MC, Kidd EA (2014) Impact of socioeconomic status and ethnic enclave on cervical cancer incidence among Hispanics and Asians in California. Physiol Behav 176(3):139–148. https://doi.org/10.1016/j.ygyno.2014.03.559.Impact
29. Von Behren J, Abrahão R, Goldberg D, Gomez SL, Setiawan VW, Cheng I (2018) The influence of neighborhood socioeconomic status and ethnic enclave on endometrial cancer mortality among Hispanics and Asian Americans/Pacific Islanders in California. Physiol Behav 176(1):139–148. https://doi.org/10.1007/s10552-018-1063-7.The